Exercise Sheet 00

Day One at Brightwell

House rules.

- Write every file yourself, in the folder `my_work/`. **Type** the code; don't copy-paste it. That's how your fingers learn.
- Run each file from inside the lesson folder: `python my_work/hello.py`
- Compare what you get with the **expected output**. Same output means the exercise is done.
- Error message? Good. Read the last line first (*what*), then find the line number (*where*).

Exercise 1 Talk to Python

10 min

Open a terminal and type `python`. For each line, **first write down your guess**, then type it and press Enter.

YOU TYPE	YOUR GUESS
`32500 / 12`	
`32500 // 12`	
`32500 % 12`	
`2 ** 10`	
`"pay" + "slip"`	
`"ha" * 3`	
`len("Brightwell Umbrellas")`	
`"5" + 3`	

Type `exit()` to leave. Which result surprised you most? Write it in your pull request.

Exercise 2 Your first script

5 min

Create the file `my_work/hello.py`. It should print two lines: a greeting, and what £32,500 a year is per month (let Python do the division).

EXPECTED OUTPUT

```
Hello from Python!
A salary of £32,500 a year is 2708.3333333333335 a month
```

Hint: `print("text", 32500 / 12, "more text")` prints several things separated by spaces.

Exercise 3 The staff directory

10 min

> **From Priya:** *"Need your details for the staff directory. Our directory is a Python script now, apparently."*

(a) Create `my_work/about_me.py`. At the top, store three things in **variables**: your name, your favourite Excel function, and one boring task you'd love to automate.

(b) Below that, print three lines using only the variables (not the text itself):

```
Hi, I'm Jen!
My favourite Excel function is XLOOKUP
I want to automate: the monthly pay gap tabs
```

(c) Change your name variable to `"Biscuit"` and run it again. How many lines did you have to change? That's why variables exist. (Change it back before Biscuit gets ideas.)

Exercise 4 Dave's Leftovers #0 *15 min*

Open `bughunt/hello_bug.py`, Dave's welcome script from 2019. It has never run. There are **three** mistakes, and Python shows you one at a time.

(a) Run `python bughunt/hello_bug.py`. Note the line number and the kind of error.

(b) Fix it, save, run again. Repeat until it prints:

```
Hello, Jen! Welcome to Brightwell.
jen@brightwell.co.uk
TEA TIME
```

(c) For each bug, write one sentence: what did Python say, and what was actually wrong?

Exercise 5 Hand it in *10 min*

```
git status
git add .
git commit -m "Lesson 00: hello, about me, Dave's script fixed"
git push
```

On GitHub, open a **pull request** from `lesson-00` into `main`. In the description, add your answers to Exercise 1 (the surprise) and Exercise 4(c) (the three bugs).